

Tanvir Arafin

Department of Cyber Security Engineering
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Academic Degrees

University of Maryland <i>Ph.D., Electrical Engineering</i>	College Park, MD 2018
University of Maryland <i>M.Sc., Electrical Engineering</i>	College Park, MD 2016
Bangladesh University of Engineering & Technology <i>B.Sc., Electrical & Electronic Engineering</i>	Dhaka, Bangladesh 2011

Academic Positions

George Mason University <i>Assistant Professor (Tenure-Track)</i> Department of Cyber Security Engineering <i>Affiliate Faculty, Center of Excellence in C5I</i>	Fairfax, VA 2022 – present Jan. 2023 – present
Morgan State University <i>Assistant Professor (Tenure-Track)</i> Department of Electrical and Computer Engineering <i>Assistant Director, Cybersecurity Assurance and Policy (CAP) Center</i>	Baltimore, MD 2019 – 2022 Aug. 2019 – May 2022
Bangladesh University of Engineering & Technology <i>Lecturer</i> Department of Electrical and Electronic Engineering	Dhaka, Bangladesh 2011 – 2012

Industrial, Consulting, and Summer Positions

Bloomberg <i>Software Engineer</i>	New York, NY 2018 – 2019
Cyber Innovation Group, Philips <i>Vulnerability Research Intern</i>	Andover, MA 2016 – 2017
Security & Privacy Group, Bosch <i>Research Intern</i>	Pittsburgh, PA 2016

Honors, Awards, & Professional Recognitions

Departmental Nominee for the Beck Family Medal for Excellence in Research & Scholarship

Department of Cyber Security Engineering, George Mason University 2026

Best Poster/Demo Award

ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec) 2025

Best Hardware Demo Award, 2nd Place

IEEE International Symposium on Hardware Oriented Security and Trust (HOST) 2024

Departmental Nominee for the Dean's Award for Excellence in Service

Department of Cyber Security Engineering, George Mason University 2024

Thank-a-Teacher Recognition

Stearns Center for Teaching and Learning, George Mason University 2024, 2023

Featured Paper of the Month

IEEE Transactions on Computers 2022

Best Paper Award

IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST) 2018

Best Paper Nominee

ACM Great Lakes Symposium on VLSI (GLSVLSI) 2017

Goldhaber Travel Grant

University of Maryland, College Park 2015

Distinguished Graduate Fellowship

A. James Clark School of Engineering, University of Maryland 2012

Student Honors and Awards under My Supervision

DAC Young Fellow

63rd Design Automation Conference (DAC), Ph.D. Student: Yanze Wu 2026
George Mason University

Third Place

Student Research Competition

International Conference on Parallel Architectures and Compilation Techniques (PACT),
Ph.D. Student: Yanze Wu 2025
George Mason University

Best Paper Award

8th Annual Sage Memorial Capstone Design Competition, Undergraduate Students: Justin Rockwell,
Max Bedewi, Mack Bartsch, Bisesh Acharya, Craig Kimbal 2025
George Mason University

Daniel R. Bannister PhD Retention Fellowship

College of Engineering & Computing, Ph.D. Student: Yanze Wu 2024
George Mason University

First Place

National Transportation Cybersecurity Competition, Undergraduate Student: Finn Schafer 2024
George Mason University

Second Place

National Transportation Cybersecurity Competition, Undergraduate Students: Noor Mohammed, Sonia Shaukat, Caleb Hughston, Kayleigh Batchos 2024
George Mason University

Third Place

National Transportation Cybersecurity Competition, Undergraduate Student: Navraj Singh Gill 2024
George Mason University

Best Paper Award

7th Annual Sage Memorial Capstone Design Competition, Undergraduate Students: Blake Todorowski, Harris Laing, Michael Fox, Rohit Engala 2024
George Mason University

Publications

I have authored/co-authored **5 book chapters, 8 refereed journal articles, and 34 conference publications and proceedings contributions**. My work has appeared in venues including **DAC, DATE, HOST, PACT, ICCAD, IEEE Transactions on Computers, and IEEE Transactions on Very Large Scale Integration (VLSI) Systems**.

Google Scholar: 702 citations; h-index: 14; i10-index: 21

Metrics retrieved Aug. 9, 2026.

Authors who were students under my supervision at the time of the work are underlined; Ph.D. students are marked with an asterisk (); my name appears in **bold**.*

Publications Since Joining George Mason University (2022–Present).....

Since joining George Mason University, I have authored/co-authored **19 scholarly publications and conference contributions**: 1 book chapter, 3 refereed journal articles, 9 refereed conference/workshop papers, 3 GOMACTech conference digest papers, 2 refereed poster/demonstration papers, and 1 late-breaking result. My Mason research program increasingly features student-led scholarship, including multiple publications led by Ph.D. students under my supervision.

Book Chapters.....

- [1] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2025. "Attack Detection and Countermeasures at Edge Devices". In: *Smart Cyber-Physical Power Systems*. John Wiley and Sons, Ltd. Chap. 20, pp. 539–553. doi: 10.1002/9781394191529.ch20.

Refereed Journal Articles.....

- [1] Wu, Yanze* and **Arafin, Md Tanvir**. 2026. "ArKANe: Accelerating Kolmogorov Arnold Networks on Reconfigurable Spatial Architectures". In: *IEEE Embedded Systems Letters*. **ESL '26** 18.1, pp. 44–47. doi: 10.1109/LES.2025.3567918. [**Impact Factor = 2**].

- [2] Lu, Z., Wang, X., **Arafin, Md Tanvir**, Yang, H., Liu, Z., Zhang, J., and Qu, G. 2024. "An RRAM-Based Computing-in-Memory Architecture and Its Application in Accelerating Transformer Inference". In: *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*. **TVLSI '24** 32.03, pp. 485–496. doi: 10.1109/TVLSI.2023.3345651. [**Impact Factor = 3.1**].
- [3] Pan, Yuqian, Lu, Zhaojun, Zhang, Haichun, Zhang, Haoming, **Arafin, Md Tanvir**, Liu, Zhenglin, and Qu, Gang. 2023. "ADLPT: Improving 3D NAND Flash Memory Reliability by Adaptive Lifetime Prediction Techniques". In: *IEEE Trans. Computers*. **TC '23** 72.6, pp. 1525–1538. doi: 10.1109/TC.2022.3214115. [**Impact Factor = 3.8**].

Submitted Journal Articles under Review.....

- [4] Van Pelt, Ethan, Ghosh, Suhrid*, Singh, Pratham, Vadlamani, Nishanth, and **Arafin, Md Tanvir**. "MAGE: Multi-Scalar Multiplication Acceleration via GLV Enhancements for Edge GPUs". In: *IEEE Transactions on Computing*. **TC '26**.
- [5] Wu, Yanze* and **Arafin, Md Tanvir**. "MuZK: An Efficient Multiplication Reduction Accelerator for Zero-Knowledge Proofs at the Edge". In: *IEEE Transactions on Computing*. **TC '26**.
- [6] Wu, Yanze* and **Arafin, Md Tanvir**. "SpatialSLH: A Compact XMSS Accelerator for Post-Quantum SLH-DSA on Spatial Architecture". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*. **TCAD '26**.

Refereed Conference and Workshop Papers.....

- [1] Fan, Andrew, Wu, Yanze*, Han, Harry, and **Arafin, Md Tanvir**. 2026. "GPU Acceleration of the Sum-Check Protocol Over Towers of Binary Fields for Verifiable Computing". In: *Proceedings of the 2026 Design, Automation and Test in Europe Conference*. **DATE '26**. Verona, Italy: IEEE/ACM, pp. 1–7. doi: 10.23919/DATE69613.2026.11539722. [**Acceptance Rate = 24.63%**].
- [2] Schafer, Finn, Lam, Ryan, Kumar, Siddarth, Tram, Jimmy, and **Arafin, Md Tanvir**. 2026. "LiDAR Spoofing for Compromising SLAM on ROS2 Using Statistical Beam Adjustment and Temporal Modeling". In: *2026 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*. **HOST '26**. Washington DC: IEEE, pp. 117–128. doi: 10.1109/HOST68814.2026.11604594. [**Acceptance Rate = 24.8%**].
- [3] Wu, Yanze*, Park, Devin, and **Arafin, Md Tanvir**. 2026. "ATTINA: Spatial Acceleration of Additive Number Theoretic Transform for Energy-Efficient zkSNARKs at the Edge". In: *To appear in the Proceedings of the ACM Design and Automation Conference*. **DAC '26**. Long Beach, CA, USA: ACM. [**Accepted, Acceptance Rate = 22.3%**].
- [4] Lebaku, Prathyush Kumar Reddy, Gao, Lu, Zhang, Yunpeng, Li, Zhixia, Liu, Yongxin, and **Arafin, Md Tanvir**. 2025. "Cybersecurity-Focused Anomaly Detection in Connected Autonomous Vehicles Using Machine Learning". In: *International Conference on Transportation and Development 2025*. **ICTD '25**, pp. 566–580. doi: 10.1061/9780784486191.051.
- [5] Lu, Zelin, Wang, Yiting, Qu, Gang, and **Arafin, Md Tanvir**. 2025. "Auto-profiling Attack on NTT Hardware Implementations". In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE, pp. 1–4. doi: 10.1109/AsianHOST68425.2025.1137. [**Acceptance Rate = 47.17%**].
- [6] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. "Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures". In: *Proceedings of the 2025 International Conference on Parallel Architectures and Compilation Techniques*. **PACT '25**.

Irvine, California, USA: ACM, pp. 270–280. doi: 10.1109/PACT65351.2025.00033. [**Acceptance Rate = 27.87%**].

- [7] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “SHAFI: Securing Hash-Based Post-Quantum Cryptography from Hardware Fault Injection Attacks”. In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE, pp. 1–4. doi: 10.1109/AsianHOST68425.2025.11370367. [**Acceptance Rate = 47.16%**].
- [8] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “THENA: Accelerating Torus Fully Homomorphic Encryption on Energy-Efficient Heterogeneous Architecture”. In: *Proceedings of the 43rd IEEE International Conference on Computer Design*. **ICCD '25**. Dallas, TX, USA: IEEE, pp. 618–625. doi: 10.1109/ICCD65941.2025.00094. [**Acceptance Rate = 25.5%**].
- [9] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2024. “SANTA: A Spatial Accelerator Design for Efficient Number Theoretic Transform (NTT) on Heterogeneous System-on-Chips”. In: *Proceedings of the 2024 Workshop on Attacks and Solutions in Hardware Security*. **CCS-ASHES '24**. Salt Lake City, UT, USA: ACM, pp. 2–10. doi: 10.1145/3689939.3695781. [**Acceptance Rate = 32.26%**].

Refereed Posters and Demonstrations.....

- [10] Somavarapu, Sreenithya, Chaudhari, Harshita, Aidlaid, Nour El Houda, Adcharyavivit, Nona, Dib, Qais, Hossain, Moinul, Shah, Vijay K, and **Arafin, Md Tanvir**. 2025. “DEMO: Radio Unit Activity Fingerprinting through Electromagnetic Side-Channel Analysis in O-RAN Networks”. In: *18th ACM Conference on Security and Privacy in Wireless and Mobile Networks*. **WiSec '25**, pp. 304–305. doi: 10.1145/3734477.3736152. [**Best Poster/Demo Award, 1/14 submissions**].
- [11] Todorowski, Blake E, Fox, Michael Lane, Laing, Harris E, Gaddam, Kirthan, Mian, Anosh, Ferrari, Jair, and **Arafin, Md Tanvir**. 2024. “Poster: Address Resolution Protocol Based Attacks for Multi-Robot Systems”. In: *2024 IEEE International Conference on Mobility: Operations, Services, and Technologies*. **MOST '24**. IEEE Computer Society, pp. 281–282. doi: 10.1109/MOST60774.2024.00041.

Late-Breaking Results.....

- [12] Wu, Yanze* and **Arafin, Md Tanvir**. 2024. “Ising Model Processors on a Spatial Computing Architecture”. In: *2024 IEEE 67th International Midwest Symposium on Circuits and Systems*. **MWSCAS '24**. IEEE, pp. 1383–1387. doi: 10.1109/MWSCAS60917.2024.10658926. [**Late Breaking Result**].

GOMACTech Conference Digest Papers.....

- [13] **Arafin, Md Tanvir**. 2025. “Large Language Models for Secure Microelectronics: Challenges and Opportunities”. In: *Government Microcircuit Applications and Critical Technology Conference (GOMACTech 2025) Digest*. 4 pages, Oral Presentation, Session 49.4. Pasadena, CA, USA.
- [14] **Arafin, Md Tanvir**, Wu, Yanze*, Matsumoto, Shou, and Costa, Paulo. 2025. “Explainable Probabilistic Models on Coarse-Grained Reconfigurable Architectures”. In: *Government Microcircuit Applications and Critical Technology Conference (GOMACTech 2025) Digest*. 4 pages, Poster Presentation, P.19. Pasadena, CA, USA.

- [15] Wang, Yiting, Lu, Zelin, Qu, Gang, and **Arafin, Md Tanvir**. 2025. "Side-Channel Vulnerabilities in the FPGA Implementations of Lattice-Based Post-Quantum Cryptographic Schemes". In: *Government Microcircuit Applications and Critical Technology Conference (GOMACTech 2025) Digest*. 4 pages, Oral Presentation, Session 53.1. Pasadena, CA, USA.

Submitted Conference Articles under Review.....

- [16] Chen, Wangxinlei* and **Arafin, Md Tanvir**. "SLAM-SimBA: Simple Black-box Attack on Graph-based Simultaneous Localization and Mapping". In: *Network and Distributed System Security (NDSS) Symposium. NDSS '27*.

Patent Applications.....

- [1] Wu, Yanze* and **Arafin, Md Tanvir**. 2026. *Parallel Hash Chain Computation and Tree Compression on Reconfigurable Multi-Tile Arrays*. Provisional patent application under consideration.

Publications Prior to Joining George Mason University.....

Book Chapters.....

- [1] **Arafin, Md Tanvir**, Xu, Qian, and Qu, Gang. 2022. "Voltage Overscaling Techniques for Security Applications". In: *Approximate Computing*. Springer, pp. 287–311. doi: 10.1007/978-3-030-98347-5_12.
- [2] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2022. "Approximation on Data Flow Graph Execution for Energy Efficiency". In: *Approximate Computing*. Springer, pp. 207–232. doi: 10.1007/978-3-030-98347-5_9.
- [3] **Arafin, Md Tanvir** and Qu, Gang. 2021. "Hardware-Based Authentication Applications". In: *Authentication of Embedded Devices*. Springer, pp. 145–181. doi: 10.1007/978-3-030-60769-2_6.
- [4] **Arafin, Md Tanvir** and Qu, Gang. 2017. "Memristor-Based Security". In: *Security Opportunities in Nano Devices and Emerging Technologies*. CRC Press, pp. 55–72. doi: 10.1201/9781315265056-4.

Refereed Journal Articles.....

- [1] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2022. "An Approximate Memory Based Defense Against Model Inversion Attacks to Neural Networks". In: *IEEE Transactions on Emerging Topics in Computing*. TETC '22 10.4, pp. 1733–1745. doi: 10.1109/TETC.2022.3179980. [Impact Factor = 6.0].
- [2] Zhang, Jiliang, Shen, Chaoqun, Su, Haihan, **Arafin, Md Tanvir**, and Qu, Gang. 2022. "Voltage Over-Scaling-Based Lightweight Authentication for IoT Security". In: *IEEE Transactions on Computers*. TC '22, pp. 323–336. doi: 10.1109/TC.2021.3049543. [Impact Factor = 3.8, Featured Paper of the Month, February 2022].
- [3] **Arafin, Md Tanvir** and Qu, Gang. 2018. "Memristors for Secret Sharing-Based Lightweight Authentication". In: *IEEE Transactions on Very Large Scale Integration Systems*. TVLSI '18 26.12, pp. 2671–2683. doi: 10.1109/TVLSI.2018.2823714. [Impact Factor = 3.1].
- [4] Gao, Mingze, Wang, Qian, **Arafin, Md Tanvir**, Lyu, Yongqiang, and Qu, Gang. 2017. "Approximate Computing for Low Power and Security in the Internet of Things". In: *IEEE Computer*. TC '17 50.6, pp. 27–34. doi: 10.1109/MC.2017.176. [Impact Factor = 3.8].
- [5] **Arafin, Md Tanvir**, Islam, Nazifah, Roy, Sourav, and Islam, Saiful. 2012. "Performance Optimization for Terahertz Quantum Cascade Laser at Higher Temperature Using Genetic Algorithm". In: *Optical and Quantum Electronics*. OQEL '12 44.15, pp. 701–715. doi: 10.1007/s11082-012-9590-z. [Impact Factor = 4].

Refereed Conference and Workshop Papers.....

- [1] Wang, Shuangbao Paul, **Arafin, Md Tanvir**, Osuagwu, Onyema, and Wandji, Ketchiozo. 2022. "Cyber Threat Analysis and Trustworthy Artificial Intelligence". In: *6th International Conference on Cryptography, Security and Privacy, CSP 2022, Tianjin, China, January 14-16, 2022*. IEEE, pp. 86–90. doi: 10.1109/CSP55486.2022.00024.
- [2] **Arafin, Md Tanvir** and Kornegay, Kevin. 2021. "Attack Detection and Countermeasures for Autonomous Navigation". In: *2021 55th IEEE Annual Conference on Information Sciences and Systems. CISS '21*. IEEE, pp. 1–6. doi: 10.1109/CISS50987.2021.9400224.
- [3] Kornegay, Michel A., **Arafin, Md Tanvir**, and Kornegay, Kevin. July 2021. "Engaging Underrepresented Students in Cybersecurity using Capture-the-Flag(CTF) Competitions (Experience)". In: *2021 ASEE Virtual Annual Conference Content Access. ASEE '21*. 7 pages. Virtual Conference: ASEE Conferences. doi: 10.18260/1-2--37048.
- [4] Lu, Zhaojun, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "RIME: A Scalable and Energy-Efficient Processing-In-Memory Architecture for Floating-Point Operations". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference. ASP-DAC '21*. IEEE, pp. 120–125. doi: 10.1145/3394885.3431524. [**Acceptance Rate = 30%**].
- [5] Gao, Jia-Bao, Wang, Jian, **Arafin, Md Tanvir**, and Lai, Jin-Mei. 2020. "FABLE-DTS: Hardware-Software Co-Design of a Fast and Stable Data Transmission System for FPGAs". In: *33rd IEEE International System-on-Chip Conference, SoCC 2020, Las Vegas, NV, USA, September 8-11, 2020*. IEEE, pp. 207–212. doi: 10.1109/S0CC49529.2020.9524764. [**Acceptance Rate = 30.1%**].
- [6] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2020. "MIDAS: Model Inversion Defenses Using an Approximate Memory System". In: *2020 IEEE Asian Hardware Oriented Security and Trust Symposium. AsianHOST '20*. IEEE, pp. 1–4. doi: 10.1109/AsianHOST51057.2020.9358254. [**Acceptance Rate = 27%**].
- [7] Yimer, Tsion, **Arafin, Md Tanvir**, and Kornegay, Kevin T. 2020. "Securing Industrial Control Systems Using Physical Device Fingerprinting". In: *7th International Conference on Internet of Things: Systems, Management and Security, IOTSMS 2020, Virtual Event, France, December 14-16, 2020*. Ed. by Larbi Boubchir, Elhadj Benkhelifa, Yaser Jararweh, and Imad Saleh. IEEE, pp. 1–6. doi: 10.1109/IOTSMS52051.2020.9340160.
- [8] **Arafin, Md Tanvir**, Shen, Haoting, Tehranipoor, Mark M, and Qu, Gang. 2019. "LPN-based Device Authentication Using Resistive Memory". In: *Proceedings of the 2019 ACM Great Lakes Symposium on VLSI. GLSVLSI '19*, pp. 9–14. doi: 10.1145/3299874.3317970. [**Acceptance Rate = 27%**].
- [9] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Guajardo, Jorge. 2018. "Probing Attacks on Physical Layer Key Agreement for Automotive Controller Area Networks". In: *2018 IEEE Asian Hardware Oriented Security and Trust Symposium. AsianHOST '18*. IEEE, pp. 7–12. doi: 10.1109/AsianHOST.2018.8607166. [**Best Paper Award**].
- [10] **Arafin, Md Tanvir**, Anand, Dhananjay, and Qu, Gang. 2017. "A Low-Cost GPS Spoofing Detector Design for Internet of Things (IoT) Applications". In: *Proceedings of the 2017 ACM Great Lakes Symposium on VLSI 2017. GLSVLSI '17*, pp. 161–166. doi: 10.1145/3060403.3060455. [**Best Paper Nominee, Acceptance Rate = 24%**].

- [11] **Arafin, Md Tanvir**, Gao, Mingze, and Qu, Gang. 2017. "VOLtA: Voltage Over-Scaling Based Lightweight Authentication for IoT Applications". In: *2017 22nd IEEE/ACM Asia and South Pacific Design Automation Conference. ASP-DAC '17*. IEEE, pp. 336–341. doi: 10.1109/ASPDAC.2017.7858345. [Acceptance Rate = 30%].
- [12] **Arafin, Md Tanvir**, Stanley, Andrew, and Sharma, Praveen. 2017. "Hardware-based Anti Counterfeiting Techniques for Safeguarding Supply Chain Integrity". In: *2017 IEEE International Symposium on Circuits and Systems. ISCAS '17*. IEEE, pp. 1–4. doi: 10.1109/ISCAS.2017.8050605.
- [13] **Arafin, Md Tanvir**, Anand, DM, and Qu, Gang. 2016. "Detecting GNSS Spoofing using a Network of Hardware Oscillators". In: *Proceedings of the 47th Annual Precise Time and Time Interval Systems and Applications Meeting (PTTI)*, pp. 74–79. doi: 10.33012/2016.13135.
- [14] **Arafin, Md Tanvir** and Qu, Gang. 2016. "Secret Sharing and Multi-User Authentication: From Visual Cryptography to RRAM Circuits". In: *Proceedings of the 26th ACM Great Lakes Symposium on VLSI. GLSVLSI '16*, pp. 169–174. doi: 10.1145/2902961.2903039. [Acceptance Rate = 25%].
- [15] **Arafin, Md Tanvir**, Dunbar, Carson, Qu, Gang, McDonald, N, and Yan, L. 2015. "A Survey on Memristor Modeling and Security Applications". In: *IEEE International Symposium on Quality Electronic Design. ISQED '15*. IEEE, pp. 440–447. doi: 10.1109/ISQED.2015.7085466.
- [16] **Arafin, Md Tanvir** and Qu, Gang. 2015. "RRAM Based Lightweight User Authentication". In: *IEEE/ACM international conference on Computer-Aided Design. ICCAD '15*. IEEE, pp. 139–145. doi: 10.1109/ICCAD.2015.7372561. [Acceptance Rate = 26%].

Invited Conference Papers.....

- [17] **Arafin, Md Tanvir**. 2022. "Computation-in-Memory Accelerators for Secure Graph Database: Opportunities and Challenges". In: *27th Asia and South Pacific Design Automation Conference, Taipei, Taiwan, January 17-20, 2022. ASP-DAC '22*. IEEE, pp. 31–36. doi: 10.1109/ASP-DAC52403.2022.9712500.
- [18] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "Security of Neural Networks from Hardware Perspective: A Survey and Beyond". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference. ASP-DAC '21*. IEEE, pp. 449–454. doi: 10.1145/3394885.3431639.
- [19] **Arafin, Md Tanvir** and Lu, Zhaojun. 2020. "Security Challenges of Processing-in-Memory Systems". In: *Proceedings of the 2020 ACM Great Lakes Symposium on VLSI (GLSVLSI). GLSVLSI '20*, pp. 229–234. doi: 10.1145/3386263.3411365.

Patents.....

- [1] **Arafin, Md Tanvir** and Kornegay, Kevin. Oct. 2025. *Attack Detection and Countermeasures for Autonomous Navigation*. US Patent 12,438,890.
- [2] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Mar. 2021. *Method to Mitigate Voltage Based Attacks on Key Agreement Over Controller Area Network (CAN)*. US Patent 10,958,680.
- [3] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Feb. 2020. *Method to Mitigate Transients Based Attacks on Key Agreement Schemes Over Controller Area Network*. US Patent 10,554,241.

Ph.D. Thesis.....

- [1] **Arafin, Md Tanvir**. 2018. "Hardware-Based Authentication for the Internet of Things". PhD thesis. University of Maryland, College Park. doi: 10.13016/M2HH6C88R.

Awarded Grants, Contracts, & Donations

PI/Lead PI on 8 and Co-PI on 7 externally funded research, education, and workforce-development projects and programs. Aggregate project value: **\$10,137,589**; individual investigator share: **\$2,223,477**.

At George Mason University.....

Aggregate Project Value: \$6,477,964

Awarded to Mason: \$5,327,964

My Share: \$1,760,977

Role: PI/Lead PI on 7; Co-PI on 3

[1] National Science Foundation [Award #2623105] 2026 – 2029

Project: CyberAI Innovation: Modernizing IoT/Edge Cybersecurity Curricula in the Embodied AI Era with Experiential Learning and Research Designs

Role: **Co-PI**

Award to Mason: \$500,000; My Share: \$200,000

PI: Jianli Pan (GMU)

[2] Virginia Innovation Partnership Authority [Award #N-1Q27-004] 2026 – 2027

Project: GAIA: GPU Microarchitecture-Based Power Attacks and Countermeasures for AI Data Centers

Role: **PI**

Award to Mason: \$100,000; My Share: \$50,000

Co-PI: Khaled Khasawneh (GMU)

[3] Virginia Innovation Partnership Authority [Award #N-2Q26-007] 2026 – 2027

Project: CASPER: Cyber-Analog Sensing for Protecting Critical Energy Infrastructure

Role: **PI**

Prime Award to Mason: \$100,000; Subaward to UVA: \$50,000; My Share: \$50,000

Co-PI: Qiaochu Zhang (UVA)

[4] 4-VA, State of Virginia 2026 – 2027

Project: ANCHOR: Analog–Cyber Sensing for Secure Energy Infrastructure

Role: **Co-PI**

Total Project Award: \$25,000; Award to Mason: \$5,000; My Share: \$5,000

PI: Qiaochu Zhang (UVA); Award to UVA: \$20,000

[5] National Science Foundation [Award #2523805] 2025 – 2028

Project: Collaborative Research: CISE: Crosscutting Small: SaTC: SAGE: Secure Accelerators for Next-Generation Foundation Models

Role: **PI**

Total Project Award: \$600,000; Award to Mason: \$275,000; My Share: \$275,000

Lead PI: Wenjie Che (Howard University)

[6] National Science Foundation [Award #2519390] 2025 – 2028
Project: Collaborative Research: CyberSTAR: CyberTraining for Secure Transportation and Reliable Autonomy
Role: Lead PI
Total Project Award: \$500,000; Award to Mason: \$200,000; My Share: \$200,000
Other PIs: Lu Gao (University of Houston); Qian Wang (UC Merced)

[7] National Science Foundation [Award #2438599] 2025 – 2029
Project: CyberCorps Scholarship for Service: EAGLE: Empowering American Government Leadership in Cybersecurity through Education
Role: Co-PI
Award to Mason: \$3,952,971; My Share: \$790,594
PI: Kun Sun (GMU); Co-PIs: Paulo Costa (GMU), Jianli Pan (GMU), Peggy Brouse (GMU)

[8] Virginia Innovation Partnership Authority [Award #N3Q24003] 2024 – 2025
Project: Securing Chiplet-Based Semiconductor Manufacturing from Untrusted Supply Chains
Role: PI
Award to Mason: \$50,000; My Share: \$50,000

[9] Virginia Innovation Partnership Authority [Award #N-3Q24-002] 2024 – 2025
Project: Fingerprinting Technology for Enhancing 5G/NextG O-RAN Supply Chain Risk Management
Role: PI
Award to Mason: \$49,993; My Share: \$45,383
Award transferred from Vijay Shah (former PI, now at NCSU); My Share before transfer: \$0; My Share after transfer: \$45,383

[10] National Science Foundation [Award #2245156] 2022 – 2026
Project: Collaborative Research: CISE-MSI: DP: CNS: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments
Role: PI
Total Project Award: \$600,000; Award to Mason: \$95,000; My Share: \$95,000
Lead PI: Kewei Sha (UNT); Other PIs: Bin Tang (CSUDH), Lily Ma (CUNY), Pooyan Fazli (ASU)

Prior to Joining George Mason University.....

Awarded at Morgan State University: \$3,659,625
My Share: \$462,500
Role: PI on 1; Co-PI on 4

[11] Maryland Industrial Partnerships (MIPS) 2022
Project: VISPR: A Verified Instruction Secure Processor
Role: PI
Prime Award to Morgan State University: \$130,000; Industry Subaward: \$20,000; My Share: \$110,000
Industry Subrecipient: Clarity Cyber, LLC.

- [12] **National Science Foundation [Award #2042700]** 2021 – 2022
Project: CyberCorps Scholarship for Service: Secure Embedded Systems
 Role: **Co-PI**
 Award to Morgan State University: \$3,184,625; My Share: \$265,000
 PI: Kevin Kornegay (MSU); Co-PIs: Michel Kornegay (MSU), Onyema Osuagwu (MSU), Ketchiozo Wandji (MSU)
- [13] **Applied Research Laboratory for Intelligence and Security (ARLIS)** 2020 – 2021
Project: Cyber-Assessment of AI/ML Tools
 Role: **Co-PI**
 Award to Morgan State University: \$150,000; My Share: \$37,500
 PI: Kevin Kornegay (MSU)
- [14] **National Security Agency (NSA), NCAE-C Program** 2020 – 2021
Project: Secure Autonomous Navigation Under Adversarial Attacks
 Role: **Co-PI**
 Award to Morgan State University: \$150,000; My Share: \$50,000
 PI: Kevin Kornegay (MSU)
- [15] **NASA Jet Propulsion Laboratory (NASA-JPL)** 2020
Project: Specification-Based Anomaly Detection for Embedded Devices
 Role: **Co-PI**
 Award to Morgan State University: \$45,000; My Share: \$0
 PI: Kevin Kornegay (MSU); Funding supported a student researcher

Invited Talks, Workshops, & Presentations

1. **Hardware for Secure Autonomy**, *C5I Center Research Review*, George Mason University, 2026.
2. **Blueprint for the T-1000: The Next Generation of Foundation Models at the Edge**, *Virginia Technology Summit*, Micron, 2026.
3. **Hardware for Secure Autonomy**, *Innovation in Cybersecurity AI Colloquium Talk Series*, University of the District of Columbia, 2025.
4. **Ghosts in the Shell: Security Issues in Robotic Operating System**, Hardware Demonstration, *Whiskey and Widgets event for College of Engineering and Computing*, George Mason University, 2024.
5. **Hardware Security @ C5I Center**, Hardware Demonstration, *Center Connect event for College of Engineering and Computing*, George Mason University, 2024.
6. **Cyber Security Issues in Modern Robotics**, *EdgeRobot Research Seminar*, California State University - Dominguez Hills, 2023.
7. **Hardware for Secure Autonomy**, *EdgeRobot Summer Research Seminar*, California State University - Dominguez Hills, 2022.
8. **Design of Secure and Efficient Processing-In-Memory Systems for Large-Scale Applications**, Tutorial Presentation, *34th International System-on-Chip Conference (SOCC)*, 2021.
9. **Hardware Lottery and the Perils of Computer Security**, Invited Talk, Computer Science

Department, IT University of Copenhagen, Denmark, 2021.

10. **Autonomous Navigation Under Adversarial Attack**, Abstract Presentation, *49th Annual IEEE Applied Imagery Pattern Recognition (AIPR) Workshop*, 2020.
11. **Physical Unclonable Functions for Security Applications**, Invited Talk, COSC Colloquium Series, Computer Science Department, Morgan State University, 2020.

Teaching

Courses Taught.....

George Mason University

Course	Term	Enroll.	Response Rate	Student Participation (/5)	Learning Outcomes (/5)	Course Environment (/5)	Prep./ Org. (/5)
CYSE 650: Adversarial Machine Learning and Cyber Security	SP 2026	12	12/12 (100%)	4.58	4.50	4.71	4.68
CYSE 465: Transportation System Design	SP 2026	28	24/28 (85.70%)	4.60	4.28	4.23	4.36
CYSE 465: Transportation System Design	SP 2025	17	13/17 (76.47%)	4.31	4.31	4.47	4.46
CYSE 465: Transportation System Design	F 2024	12	11/12 (91.67%)	4.55	4.80	4.64	4.85
CYSE 211: Operating Systems and Lab	F 2024	48	39/48 (81.25%)	4.03	4.00	3.99	4.04
CYSE 465: Transportation System Design	SP 2024	10	9/10 (90.00%)	4.89	4.78	4.80	4.74
CYSE 465: Transportation System Design	F 2023	25	23/25 (92.00%)	4.35	4.63	4.57	4.59
CYSE 499/580: Hardware and Cyber-Physical Systems	SP 2023	37	33/37 (89.19%)	4.26	4.16	4.35	4.27
CYSE 465: Transportation System Design	F 2022	39	20/39 (51.28%)	4.13	2.65	3.10	2.48

Note: George Mason University's Student Evaluations of Teaching (SET) instrument reports competency/category means rather than separate overall "Instructor Rating" and "Course Rating" scores. All four SET competency means are reported on a 5-point scale.

Morgan State University

- o EEGR 463: Digital Electronics SP 2022, SP 2021, F 2020, SP 2020, F 2019
- o EEGR 745: Advanced Digital VLSI Design F 2021
- o EEGR 480: Introduction to Cyber Security F 2020, F 2019
- o EEGR 760: Advanced Topics in Computer Engineering SP 2020

Bangladesh University of Engineering & Technology

- o Introduction to Electrical Engineering SP 2012
- o VLSI I Laboratory SP 2012, F 2011
- o Microprocessor & Interfacing Laboratory F 2011
- o Electronics Laboratory F 2011

Program, Curriculum, and Course Development.....

- o **Ph.D. in Cybersecurity, George Mason University.** As Chair and Member of the Ph.D. Proposal Committee, served as a contributor to the development of Mason's new Ph.D. in Cybersecurity

program, including program structure, curricular requirements, and the academic proposal leading to approval by the State Council of Higher Education for Virginia (SCHEV).

- **CYSE 650: Adversarial Machine Learning and Cyber Security.** Developed the complete graduate course, including the syllabus, course content, lectures, assignments, projects, assessments, and instructional materials. The course introduces students to security and robustness challenges in modern machine-learning systems and corresponding attack and defense techniques.
- **CYSE 465: Transportation System Design.** Significantly redesigned and expanded the course through activities supported by an NSF CyberTraining award, incorporating contemporary transportation cybersecurity topics, hands-on laboratory exercises, security analysis, and experiential learning involving automotive and autonomous systems.

Teaching Innovations and Instructional Materials

- **Hands-on Transportation Cybersecurity Laboratories.** Developed and integrated experiential laboratory exercises from NSF CyberTraining Award and our research into CYSE 465: Transportation System Design, including activities in automotive-network security, CAN-bus exploitation, reverse engineering, and transportation-system attack and defense.
- **Research-Informed Instruction.** Integrated contemporary research topics and tools into undergraduate and graduate instruction, connecting course material with current research in hardware security, cyber-physical systems, autonomous systems, adversarial machine learning, and secure computing.
- **Adversarial Machine Learning Instructional Materials.** Developed original lectures, assignments, projects, demonstrations, and assessments for CYSE 650: Adversarial Machine Learning and Cyber Security, with emphasis on practical attacks, defenses, and security evaluation of machine-learning systems.
- **Experiential and Project-Based Learning.** Developed course projects and practical exercises that require students to apply cybersecurity concepts to realistic systems and tools rather than relying solely on lecture-based instruction.

Mentoring & Advisement

Ph.D. Thesis Advisement

George Mason University

- **Yanze Wu** — Ph.D. Candidate, Advanced to Candidacy.
Dissertation: “Hardware Acceleration for Post-Quantum Secure Trustworthy Computing”
Expected Graduation: Spring 2027.
- **Wangxinlei Chen** — Ph.D. Student.
Dissertation Topic: Hardware Design for Secure Autonomous Systems
Expected Graduation: Fall 2028.
- **Max Duverneuil** — Ph.D. Student.
Dissertation Topic: Microarchitecture-Based Power Attacks and Countermeasures
Expected Graduation: Spring 2030.

Undergraduate Senior Design Project Sponsor

George Mason University

- Project: Fingerprinting 5G Radios for Security Application
Students: Nour El Houda Aidlaid, Nongnapat Adchariyavivit, Harshita Chaudhari, Qais Dib, Sreenithya Somavarapu 2025
- Project: GleNN: Large Language Models for Static Reverse Engineering
Students: Justin Rockwell, Maximilian Bedewi, Craig Kimbal, Mack Bartsch, Bisesh Acharya 2025
- Project: LOCO: Secure Localization for Multi-Robot Systems in Dynamic Environments
Students: Leigh Black, Shreyas Gadiraju, Andrew Le, Logan Minto, Rahul Patel, Biraj Joshi 2025
- Project: Securing Robust Multi-Robot Systems in Dynamic Environments
Students: Blake E Todorowski; Michael Lane Fox; Harris E Laing; Kirthan Gaddam; Anosh A Mian 2024
- Project: Émpistos: Trusted Embedded Processing and Optimizations for Edge AI Security
Students: Zack Francis Wagner; AlRaheeq Al Rawas; Anay Gulati; Maximilian A Karen; Bradford Williams 2024

Doctoral Thesis Committee Member

Morgan State University

- Latha Suravasi, Khir Henderson, Greig Richmond, Edmund Smith, Tsion Yimer

Undergraduate Senior Design Project Supervisor

Morgan State University

- Maryline Ivana Happy; Jose Dominguez-Cortez; Robert Hill 2022
- Ashia McCalla; Gerald Amory 2022
- Antwaan Thomas; Faizat Kaffo 2021
- Malik Smith; Anthony Turner 2021
- Fitsum Tadasse; Reuben Macintosh 2021

External Professional Service

Grant and Proposal Review.....

- Proposal Reviewer, Commonwealth Cyber Initiative (CCI) 2025
- Report Reviewer, USDOT National Center for Transportation Cybersecurity and Resiliency 2025
- Panelist, Secure and Trustworthy Cyberspace (SaTC), National Science Foundation (NSF) 2022 – 2024
- Technical Reviewer, Maryland Industrial Partnerships (MIPS) 2021

Conference Organization and Leadership.....

- TPC Chair, Section 2: Hardware Security, IEEE Design and Automation Conference (DAC) 2025
- Publicity Chair, IEEE Asian Hardware Oriented Security and Trust Symposium 2025
- Publication Chair, IEEE Asian Hardware Oriented Security and Trust Symposium 2021 – 2024

Conference Technical Program Committees.....

- IEEE Design and Automation Conference (DAC) 2023 – 2024, 2026

IEEE International Symposium on Hardware Oriented Security and Trust (HOST)	2026
ACM Great Lakes Symposium on VLSI (GLSVLSI)	2026
IEEE International System-on-Chip Conference (SOCC)	2020 – 2021, 2023 – 2024
IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST)	2021 – 2024
IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)	2021 – 2023

Conference Session Chair.....

IEEE International Symposium on Hardware Oriented Security and Trust (HOST)	2026
IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)	2020, 2022
IEEE International System-on-Chip Conference (SOCC)	2020, 2021
IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST)	2021

Journal Reviewing.....

IEEE Transactions on Computer-Aided Design of ICs and Systems (TCAD)
IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
IEEE Transactions on Circuits and Systems I (TCAS I)
IEEE Transactions on Information Forensics and Security (TIFS)
IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)
IEEE Internet of Things Journal (IoTJ)
IEEE Network Magazine
ACM Journal on Emerging Technologies in Computing Systems (JETC)
Elsevier Integration, the VLSI Journal
Computer & Security
Springer Journal of Hardware and Systems Security

University and Departmental Service

George Mason University

Member, Departmental Tenure-Track Hiring Committee, CYSE Department	2025 – 2026
Member, Ph.D. Proposal Committee, CYSE Department	2025 – 2026
Member, Departmental Tenure-Track Hiring Committee, CYSE Department	2024 – 2025
Chair, Ph.D. Proposal Committee , CYSE Department	2023 – 2024
Member, Department Handbook Committee, CYSE Department	2023 – 2024
Member, Grievance Committee, CYSE Department	2023 – 2024
Chair, Graduate Committee , CYSE Department	SP 2023
Member, Advertisement Committee, CYSE Department	SP 2023
Member, Departmental Tenure-Track Hiring Committee, CYSE Department	2022 – 2023
Member, Departmental Term-Track Hiring Committee, CYSE Department	2022 – 2023
Chair, Colloquium Committee , CYSE Department	F 2022
Member, Graduate Committee, CYSE Department	F 2022

Morgan State University

Graduate Coordinator , ECE Department, Morgan State University	2020 – 2022
Undergraduate Coordinator , ECE Department, Morgan State University	2019 – 2020
Member, Curriculum Development Committee, Ph.D. in Secure Embedded Systems	2020

Member, Faculty Development Committee, ECE Department	2019 – 2022
Member, Cyber Defense Education (CAE-CDE) Re-designation Committee	2020, 2021
USENIX Campus Representative, Morgan State University	2020 – 2022

Professional Memberships

Member, Institute of Electrical and Electronics Engineers (IEEE)	2008 – present
Member, Association for Computing Machinery (ACM)	2025 – present
Member, ACM Emerging Interest Group on Trustworthy and Responsible Systems	2026 – present
Member, USENIX: The Advanced Computing Systems Association	2020 – 2022
Member, Sigma Xi, the Scientific Honorary Society	2019 – 2023
Student Member, IEEE Communication Society	2010 – 2015

Media Coverage

1. *Timing the perfect hardware hack*, CEC Courier, 2026.
2. *Digital drive: Protecting the future of transportation*, CEC Courier, 2025.
3. *Cyber sweep: George Mason students take top three spots in national transportation cybersecurity competition*, The George, 2025.
4. *Cybersecurity students prepare for the inaugural DistrictCon Hacker Conference*, The George, 2025.
5. *Arafin conducting research aimed at securing chiplet-based semiconductor manufacturing from untrusted supply chains*, Chiplet Marketplace, 2024.